

Douglas J. Williams

Technical Product Owner | AI-Assisted Software Delivery

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Portfolio: hire-me-one-steel.vercel.app · GitHub: github.com/djwilli69

Open to fully remote opportunities

PROFESSIONAL SUMMARY

Technical Product Owner and AI-assisted software builder with 28+ years in mission-critical network operations, product leadership, and internal-tool delivery. I define product vision, translate business needs into requirements, and direct AI-assisted development from specification through testing and debugging to a working result. At CPKC, I drove a company-wide event-analysis platform (LocoGrip) that processes billions of PTC messages per day, now the primary Event & Incident Analysis tool for 45 users across 5 teams, and independently built and run a production app tracking 58 oil & gas wells across 6 operators. I pair deep systems knowledge with practical delivery — directing AI to build in Python, SQL, React, and Supabase. Seeking a fully remote Technical Product Owner or AI Product Owner role focused on internal tools, operational applications, and business-driven AI solutions.

CORE COMPETENCIES

Product Ownership: Product vision & requirements definition · business-to-technical translation · scope & delivery ownership · operational validation · stakeholder & vendor coordination

AI-Assisted Development: Spec-driven development · directing AI coding agents (Kiro) · requirements-driven delivery · iterative testing & debugging to a working result

Software & Data (AI-directed): Python · React · Supabase/Postgres · SQL · Oracle · data ingestion · Git/GitHub · Vercel · HTML/CSS

Technical Operations: Linux & Windows server administration · virtual servers · 300+ mission-critical servers · Linux/command line · log analysis at scale (Elasticsearch · Kibana) · incident response · 24/7 mission-critical support · technical documentation (MOPs)

Domain Expertise: Positive Train Control (PTC) · telecom & network operations · SMS/MMS platforms · IMS Core · oil & gas

PROFESSIONAL EXPERIENCE

Senior PTC Applications Engineer — Canadian Pacific Kansas City (CPKC)

2022 – Present · Remote

- Own the vision and delivery of internal tools supporting Positive Train Control operations across a safety-critical rail network.
- Manage 300+ mission-critical Linux and Windows servers running the messaging and dispatching services between the Train Management Dispatching System (TMDS) and locomotives — infrastructure that underpins rail safety and on-time delivery.
- Owned the vision and delivery of LocoGrip, a company-wide event-analysis platform processing billions of PTC messages per day — now the primary Event & Incident Analysis tool used by 45 people across 5 teams.
- Automated the log-gathering and diagnosis work that previously took analysts up to an hour per event, so operational issues are isolated and investigated for resolution far faster than manual review.
- Delivered Python database-integration tools connecting SQL and Oracle data sources into custom applications and reporting workflows.
- Define requirements and target outcomes for AI-assisted, spec-driven development; direct implementation and perform iterative testing and debugging through to a working result.
- Apply deep operational knowledge to validate software behavior against real-world PTC requirements.

ATS Senior Network Engineer — T-Mobile (formerly Sprint)

Jul 1998 – Aug 2022

- Owned and operated the SMS/MMS messaging platform through a 24-year tenure spanning the Sprint–T-Mobile merger, delivering high-volume consumer messaging with 24/7/365 reliability across virtualized server infrastructure.
- Product Lead for MCO (Message Controller) handling SMS/MMS traffic; led the migration of legacy Sprint messaging onto the Replica MCO platform after the T-Mobile acquisition, coordinating the cutover on mission-critical infrastructure and consolidating onto a single network.
- Implemented a virtual SMS/MMS messaging platform with Mavenir and Rackspace; led SMS migration to the IMS Core via SIP with CDMA fallback, and internalized MMS storage/processing (Syniverse → virtualized MMSC), eliminating outside managed-services costs.
- Saved **\$1.2M** over three years by renegotiating the SPAM-filtering vendor contract while maintaining service levels.
- Authored Methods of Procedure (MOPs) and trained teams on new products during rollout; drove proactive tooling and processes to identify and resolve issues before customer impact.

SELECTED PRODUCT & SOFTWARE PROJECTS

LocoGrip — PTC Event-Analysis Platform — Python · SQL/Oracle · AI-assisted (CPKC)

Company-wide operational platform processing billions of PTC messages per day; now the primary Event & Incident Analysis tool for 45 users across 5 teams. Automates the log isolation and diagnosis that previously took up to an hour per event. Owned the vision and requirements, drove development, and validated the result through testing and debugging.

Oil & Gas Well-Tracker Application — React · Supabase/Postgres · Python · Vercel (J&T Cook Corporation, own business)

Designed and delivered a full-stack app to track working-interest investments — investment vs. revenue, payout status, operator-statement ingestion, bank reconciliation, and tax reporting. Tracks 58 working-interest wells (48 currently producing) across 6 operators, ingesting monthly operator settlement statements and reconciling roughly 170 net barrels of oil per month against bank activity. Owned the product vision and requirements; directed AI-assisted implementation and debugged through to live production deployment. Sanitized demo:

github.com/djwilli69/well-tracker-demo

Database Integration Tools — Python · SQL · Oracle · AI-assisted (CPKC)

Utilities connecting operational database sources into custom applications and reporting workflows; defined requirements, directed implementation, and tested against real operational needs.

BUSINESS OWNERSHIP

Owner — J&T Cook Corporation (Oil & Gas)

Own and operate an oil & gas business with working interests in 58 wells across 6 operators, including building its software with AI-assisted development. Bring a product owner's mindset: I understand the business goal behind the requirements and deliver tools that support operational and financial needs.

EDUCATION

Bachelor of Science, Business Management — Friends University, Wichita, KS